

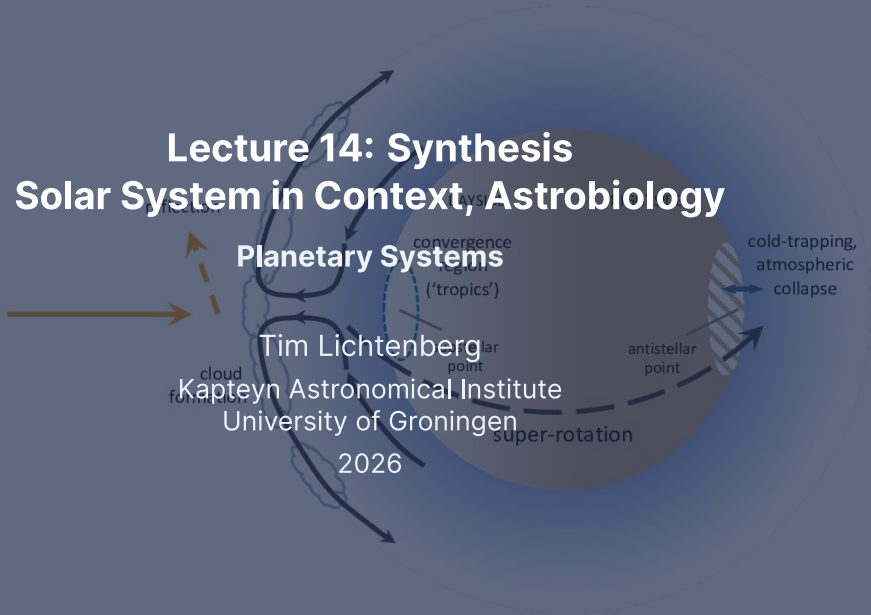
Lecture 14: Synthesis Solar System in Context, Astrobiology

Planetary Systems

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Learning Objectives

By the end of this lecture, you will be able to:

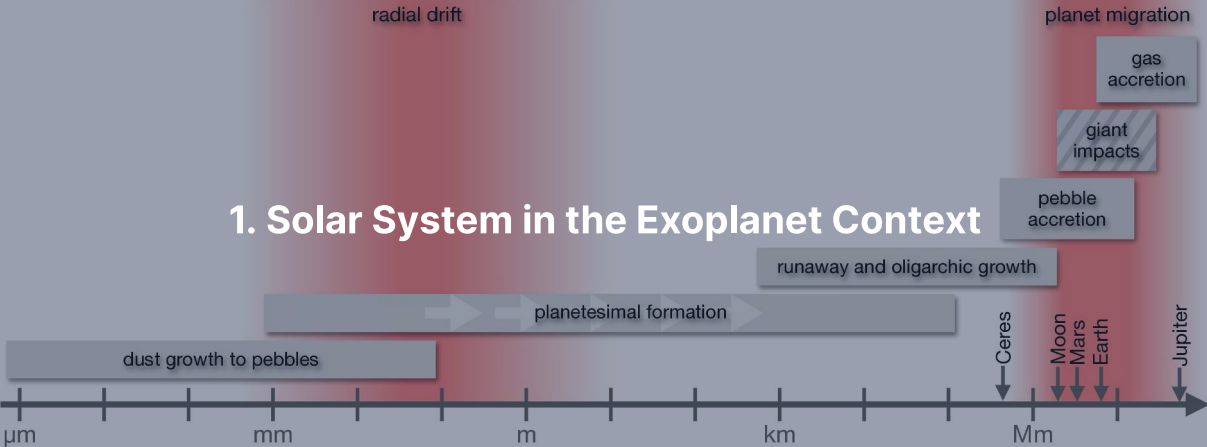
- Place our solar system within the observed exoplanet population.
- Evaluate habitability as a coupled *systems* property, not a single condition.
- Derive the classical habitable-zone boundaries from stellar luminosity and the liquid-water condition.
- Identify the most promising life-detection targets for the next decade.

Habitability is a trajectory, not a snapshot.

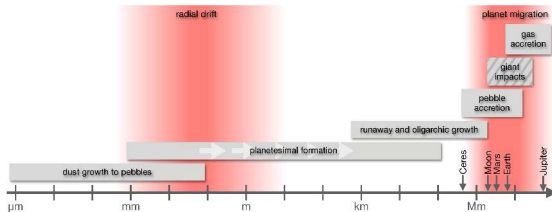
The Thread of the Course

- L1–L2: solar system + protoplanetary disks.
- L3–L4: heat, energy, differentiation, magnetospheres.
- L5–L6: atmospheres I + II.
- L7–L8: surfaces + interiors.
- L9–L12: solar system body by body.
- L13: exoplanets.
- Today: **step back**. Same physics everywhere; what does the integrated picture look like?

1. Solar System in the Exoplanet Context

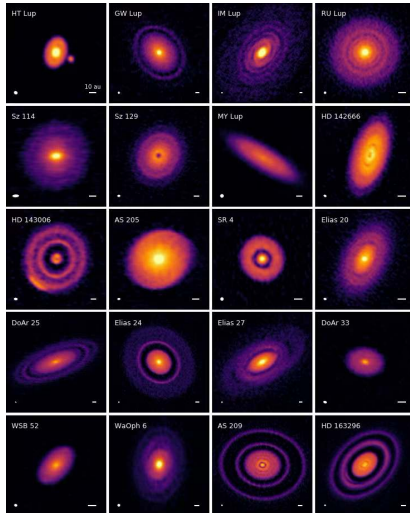


Formation Theory Meets Observation

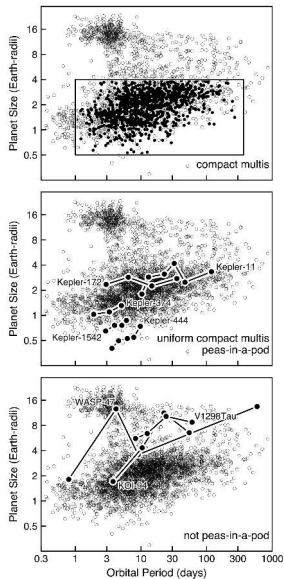


- Dust $\rightarrow$ pebbles $\rightarrow$ planetesimals (km-scale bodies) $\rightarrow$ embryos.
- Streaming instability (pebble clumping) + pebble accretion + runaway envelope accretion.
- Disk lifetime 3–5 Myr is the deadline.
- ALMA disk substructure is direct observational input.

ALMA's DSHARP Reveals Forming-Planet Signatures



- 240 GHz continuum, 20 nearby disks.
- Concentric rings, gaps, asymmetries: nearly ubiquitous.
- Most are interpreted as signatures of in-progress planet growth.
- Andrews+ 2018.

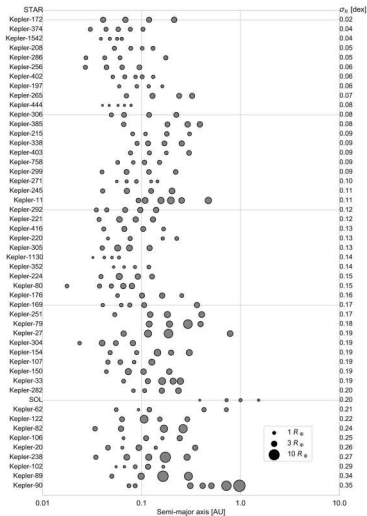


Transiting census; box and filled circles are compact multis (top), five uniform ones (middle), three non-uniform (bottom). Weiss et al. (2023, PPVII).

Where the Solar System is Atypical

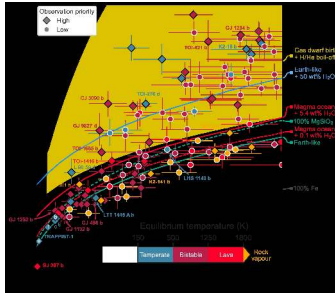
1. **No super-Earth or sub-Neptune.** The most common single class of planet, and we have none.
2. **Wide, low-eccentricity giant orbits.** Many exoplanet giants are eccentric, inclined.
3. **No hot Jupiter, no hot Neptune.** Both classes occur in $\sim 0.5\text{--}1\%$ of systems but not in ours.
4. **Irregular inner-system spacing.** Solar system terrestrials span $18\times$ in mass; not a peas-in-a-pod.

Peas-in-a-Pod: Solar System Doesn't Fit



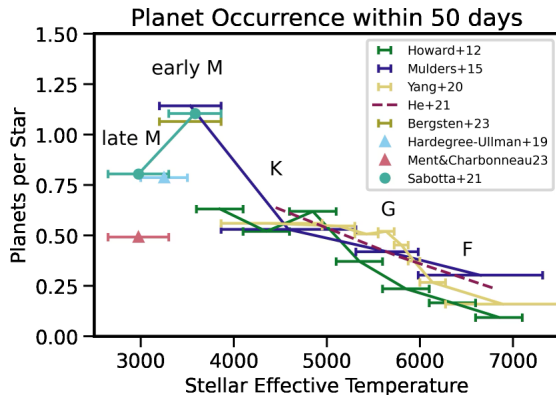
- Compact multi-systems with ≥ 4 transiting planets within 1.52 AU.
- Ranked by uniformity of planet sizes.
- Solar system: *bottom quintile* of size uniformity.
- Inner solar system did not produce the typical compact-multi architecture.

Mass-Radius: Where Each Planet Sits



- Lichtenberg 2025.
- EOS (equation-of-state) tracks: 100% Fe, Earth-like, MgSiO_3 , water-rich, magma + H_2O , gas dwarf.
- Solar system terrestrials sit on the Earth-like curve.
- Many close-in super-Earths sit there too: *stripped cores* (lost their gas envelopes).
- Larger planets carry volatile envelopes that swell their radii.

Occurrence Climbs Toward M Dwarfs

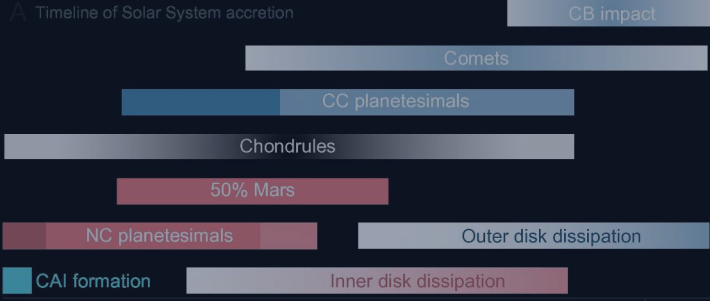


- Small planets ($1-4 R_{\oplus}$, $P < 50$ d) per star vs T_{eff} .
- Occurrence roughly 2× higher around early M dwarfs than F dwarfs.
- Trend continues into late M.
- Mulders+ 2024.
- **Earth-sized planets are common around the most numerous class of stars.**

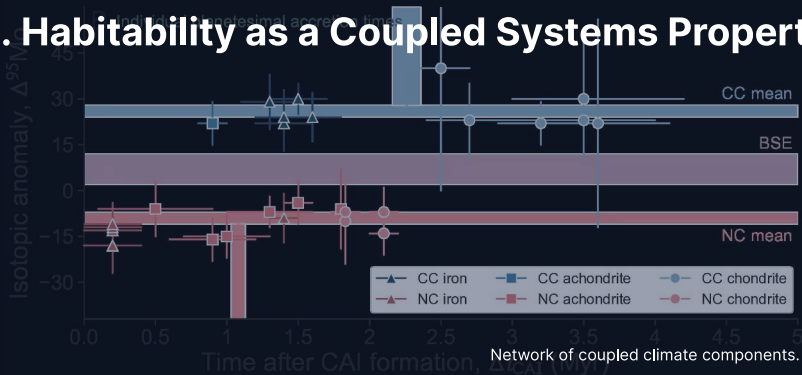
Is the Solar System Rare?

Honest answer: **not obviously, but not yet known.**

- Long-period detection floor set by survey duration. Kepler had only 4 yr.
- Earth analogues at 1 AU around G dwarfs: at the edge of current sensitivity.
- PLATO (2026) + Gaia DR4 (late 2026) will resolve.
- Confident now: inner solar system is *atypical* relative to known compact systems.
- Open: whether this is bad luck, Jupiter's fingerprint, or the typical outcome around stars we haven't sampled yet.



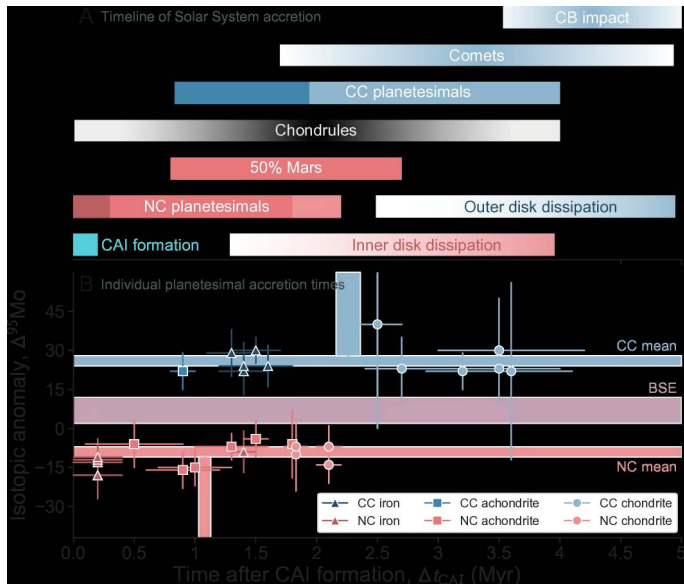
2. Habitability as a Coupled Systems Property



The Stack of Requirements

Habitability is not a checklist. It is a stack of necessary couplings:

- **Star:** stable luminosity over Gyr; not too active.
- **Orbit:** in HZ; eccentricity not too high.
- **Planet bulk:** massive enough to retain atmosphere + drive interior.
- **Atmosphere:** temperature, pressure, escape rate compatible with surface water.
- **Interior:** heat budget drives outgassing + dynamo (magnetic-field generation).
- **Surface + tectonics:** long-term volatile recycling.
- **Biosphere:** once present, modifies all of the above.

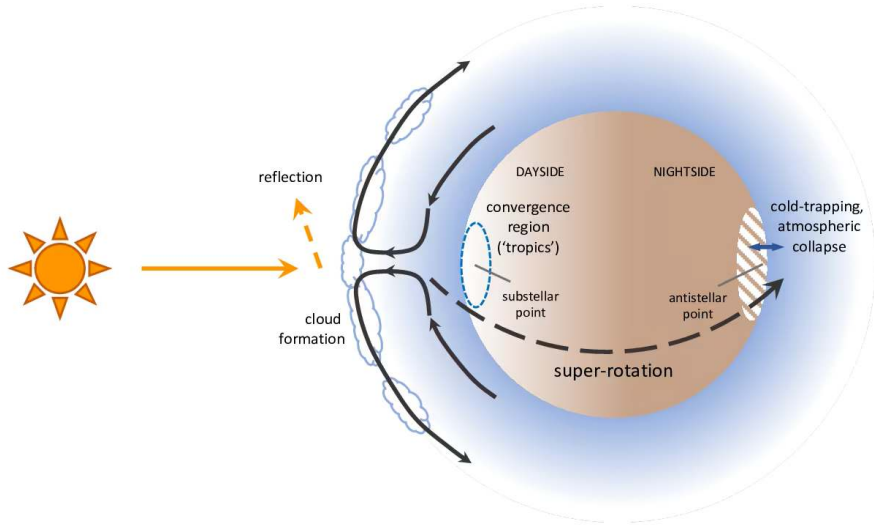


Network of coupled climate components: interior $\leftrightarrow$ atmosphere $\leftrightarrow$ surface $\leftrightarrow$ biosphere over Gyr timescales. Lichtenberg et al. (2023).

Reading the Coupling Stack

- Interior ↔ atmosphere: outgassing, volatile partitioning, magma-ocean chemistry.
- Atmosphere ↔ surface: weathering, photochemistry, redox state of near-surface volatiles.
- Surface ↔ interior: tectonic recycling, sequestration of carbonates and water.
- Star ↔ atmosphere: irradiation, XUV (X-ray + extreme-UV)-driven escape, photochemical loss.
- Biosphere (once present) feeds back on atmosphere, surface, and ocean chemistry.

Break any link, and the system can drift into a regime where surface liquid water is no longer thermodynamically possible.

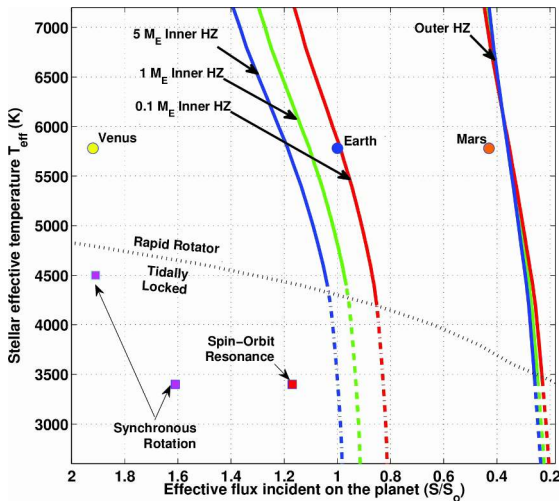


Atmospheric circulation on a tidally locked (same face toward the star) rocky planet: irradiation, clouds, super-rotation, day-night heat transport. Wordsworth & Kreidberg (2022).

Reading the Tidally Locked Circulation Diagram

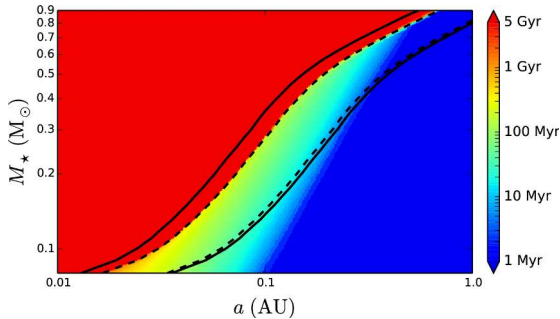
- Permanent dayside receives all stellar irradiation; partial reflection sets the planetary albedo.
- Substellar convergence region: convection lifts moisture, forms thick clouds.
- Equatorial super-rotation transports heat from dayside to nightside.
- Permanent nightside acts as a cold trap; condensable volatiles can freeze out.
- Atmospheric collapse possible when day-night transport is insufficient to keep volatiles vapour.

The HZ is Not a Line



- Inner edge: runaway greenhouse.
- Outer edge: maximum CO_2 greenhouse.
- Two planets at the same flux can have radically different climates (Earth vs Venus).
- Time-dependent: stars brighten over MS lifetime.
- HZ planet today $\neq$ habitable trajectory.

M Dwarf HZ: Pre-MS Runaway Phase



- Duration of runaway greenhouse on water-bearing planet vs $M_{\star} + a$.
- Late M dwarfs: Gyr of runaway before star reaches MS.
- Habitable-zone planet started *inside* runaway zone.
- Loses much of initial water before life could plausibly start.
- Luger & Barnes (2015).

3. Magma Oceans, Water Delivery, Tectonic Thermostat

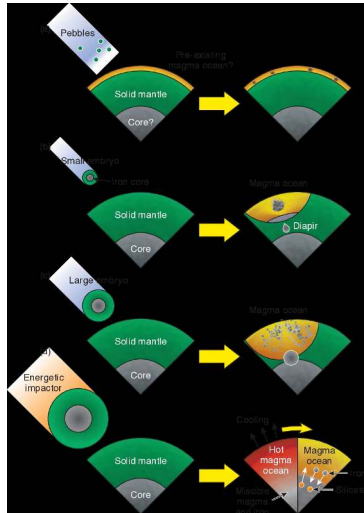


Magma-ocean differentiation. Lichtenberg et al. (2023)

Why "Delivery" Isn't the Right Frame

- Earth, Venus, Mars likely received *similar* initial volatile inventories.
- Yet present-day water inventories differ by orders of magnitude.
- Difference: **what each body did with its initial inventory.**
- Four levers (in order of importance):
 - Magma ocean (molten mantle) partitioning (core vs mantle vs steam atmosphere)
 - Magma ocean atmospheric escape
 - Mantle-atmosphere exchange over Gyr
 - Stellar evolution history

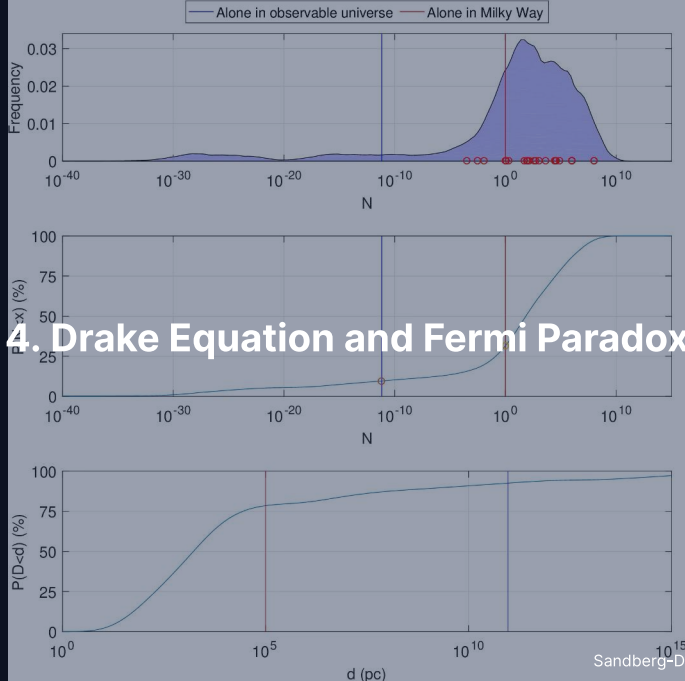
Magma Ocean Differentiation Sets the Boundary Conditions



- Initially molten silicate mantle separates from denser metal.
- Volatiles partition between mantle and steam atmosphere.
- Fraction sequestered vs outgassed depends on redox state, solidification timescale, depth.
- *Identical* starting inventories → very different end states.
- Lichtenberg et al. (2023).

Tectonic Regime: The Long-Term Thermostat

- Volcanism injects CO_2 ; rainfall + silicate weathering removes it; subduction returns carbonates to mantle.
- Negative feedback: warmer climate $\rightarrow$ faster weathering $\rightarrow$ cooler subsequent climate.
- Three conditions, all required:
 - Liquid water
 - Active volcanism
 - Tectonic recycling of carbonates back to the mantle
- Venus failure mode: surface dries $\rightarrow$ no weathering sink $\rightarrow$ CO_2 accumulates to 92 bar.
- For exoplanets: tectonic regime is *unmeasurable remotely* with current/near-term tools.



The Drake Equation

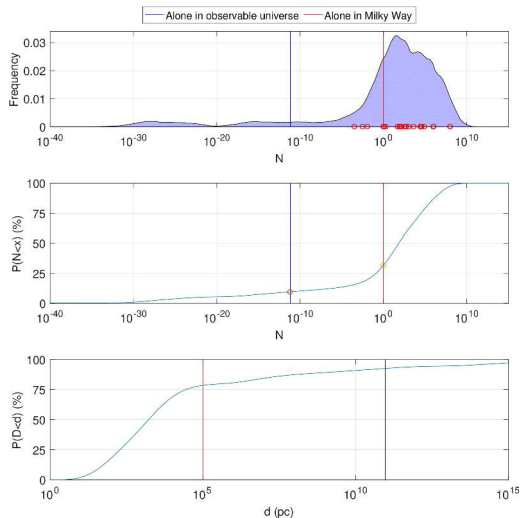
$$N = R_{\star} \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

- Only the first three factors are now measured.
 - $R_{\star}$: stellar formation rate. Astrophysics.
 - f_p : fraction with planets. Of order unity (Kepler).
 - n_e : HZ planets per system. ~ 0.1–0.6 for Sun-like (Bryson+ 2021).
- f_l, f_i, f_c, L : *not* constrained by orders of magnitude.
- Drake equation = framing tool, **not an estimator**.

Five Reasons Not to Quote a Number for N

1. Factors are not independent. Biospheres co-evolve with atmospheres.
2. Most factors span 4–10 orders of magnitude in published estimates.
3. Treats a contingent process as a steady-state pipeline.
4. Implicitly assumes uniformity over diverse planets and stars.
5. Anthropocentric selection in the only data point we have.

Sandberg+ 2018: Honest Posterior Dissolves Fermi



- Monte Carlo each Drake factor over its full published range.
- Result is *bimodal*.
- ~ 1/3 chance of being alone in the Milky Way (CDF at $N < 1$).
- ~ 10% chance of being alone in the entire observable universe (CDF at $N \ll 1$).
- Apparent silence is unsurprising once priors are honest.

Why Bimodal? Log-Sum of Broad Priors

Take the log:

$$\log_{10} N = \log_{10}(R_{\star} f_p n_e) + \log_{10} f_l + \log_{10} f_i + \log_{10} f_c + \log_{10} L$$

- Each soft factor has log-uniform prior over many decades.
- Sum of log-uniform variables = quasi-Gaussian by central limit theorem.
- Width: ~ 30 decades. Tail mass below $N = 1$ is large.
- Lesson: **broad priors on multiplicative factors generate broad and skewed product distributions.**

Fermi Paradox: A Question, Not an Answer

Possible resolutions:

- **Rare-life:** f_l very small.
- **Rare-intelligence:** f_i is the bottleneck.
- **Great filter ahead:** L short; civilisations self-terminate.
- **Detection threshold:** signals exist, are missed.
- **Zoo / quiet:** civilisations conceal themselves or use signalling we cannot recognise.

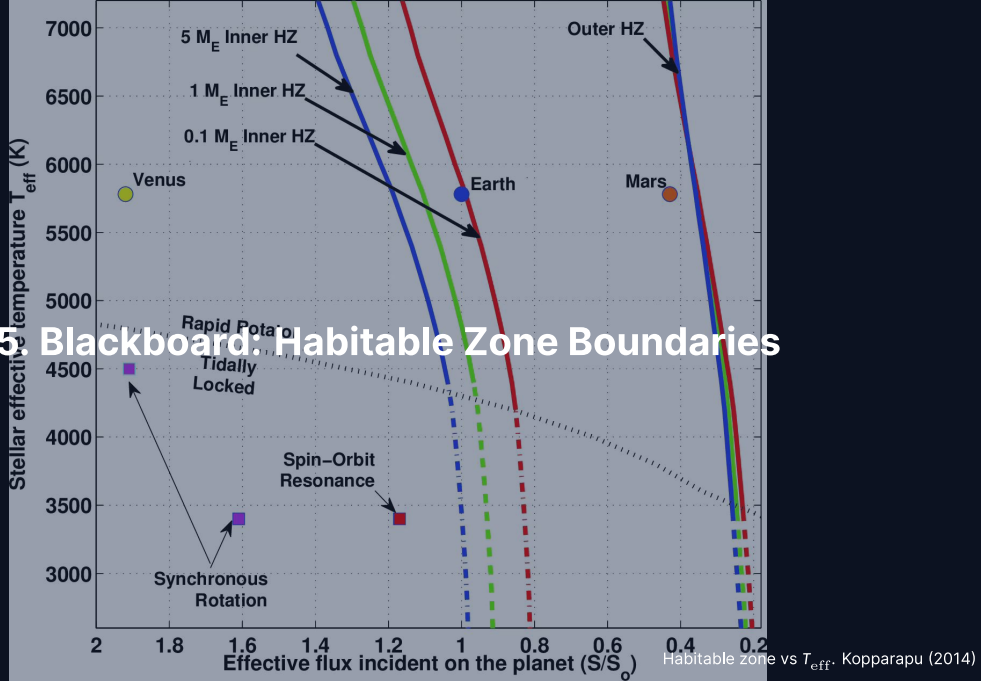
Use Fermi paradox as a check on intuition about Drake-equation factors. *Not* as evidence.

Break

End of Part 1: Solar System Context + Habitability

Part 2: Habitable Zone Derivation, Astrobiology, Life Detection

5. Blackboard: Habitable Zone Boundaries



Setup: Stellar Flux and Equilibrium Temperature

Star with bolometric luminosity $L_{\star}$, planet at distance d :

$$F_{\star}(d) = \frac{L_{\star}}{4\pi d^2}$$

Bond albedo A_B (reflected-light fraction), fast-rotating sphere, equilibrium:

$$T_{\text{eq}}(d) = \left[\frac{L_{\star}(1 - A_B)}{16\pi\sigma\epsilon d^2} \right]^{1/4}$$

For Earth: $A_B = 0.3$, $\epsilon = 1$, $L_{\star} = L_{\odot}$, $d = 1 \text{ AU} \rightarrow T_{\text{eq}} \approx 255 \text{ K}$.

Surface is 288 K because of $\sim 33 \text{ K}$ of greenhouse warming.

Effective Stellar Flux

Define $S_{\text{eff}} = \text{TOA flux at HZ boundary} / S_{\oplus} = 1361 \text{ W m}^{-2}$.

$$d = (1 \text{ AU}) \sqrt{\frac{L_{\star}/L_{\odot}}{S_{\text{eff}}}}$$

Read S_{eff} off Kopparapu+ 2013 1D radiative-convective calculation:

- Inner edge (Sun-like): $S_{\text{in,eff}} \approx 1.06$.
- Outer edge (Sun-like): $S_{\text{out,eff}} \approx 0.35$.

Inner Edge: Simpson-Nakajima Limit

In a moist atmosphere, OLR (outgoing longwave radiation) is bounded above:

$$F_{\text{OLR}}^{\text{max}} \approx 280 \text{ W m}^{-2}$$

- Once absorbed flux > this, no radiative equilibrium possible.
- Oceans driven into the atmosphere → runaway greenhouse.
- Sun-like inner edge:

$$d_{\text{in}} = (1 \text{ AU})/\sqrt{1.06} \approx 0.97 \text{ AU}$$

Venus at 0.72 AU receives $S_{\text{eff}} \approx 1.9$: *well inside* the inner edge.

Outer Edge: Maximum CO₂ Greenhouse

At low T + high p_{CO_2} , CO₂ condenses out as ice clouds.

- Maximum greenhouse warming saturates.
- No matter how much CO₂ you add, surface stays cold.
- Sun-like outer edge:

$$d_{\text{out}} = (1 \text{ AU})/\sqrt{0.35} \approx 1.69 \text{ AU}$$

Mars at 1.52 AU is just inside the outer edge.

HZ Across Stellar Types

HZ width $\propto \sqrt{L_{\star}}$.

- **G dwarf** ($L_{\odot}$): $d = 0.97\text{--}1.69$ AU. Lifetime ~ 10 Gyr.
- **K dwarf** ($0.3 L_{\odot}$): $d = 0.53\text{--}0.93$ AU. Lifetime 24 Gyr. *May be most habitable class* (Cuntz & Guinan 2016).
- **M dwarf** ($5 \times 10^{-4} L_{\odot}$, e.g. TRAPPIST-1): $d = 0.022\text{--}0.038$ AU. Inside Mercury's orbit. Lifetime 10^{12} yr but: tidally locked + Gyr pre-MS runaway.

Main-Sequence Lifetime: A One-Line Scaling

$$E_{\text{nuc}} \propto M_{\star}, L_{\star} \propto M_{\star}^{3.5}:$$

$$t_{\text{MS}} \sim \frac{E_{\text{nuc}}}{L_{\star}} \propto M_{\star}^{-2.5}$$

Normalising on Sun ($t_{\text{MS},\odot} \approx 10$ Gyr):

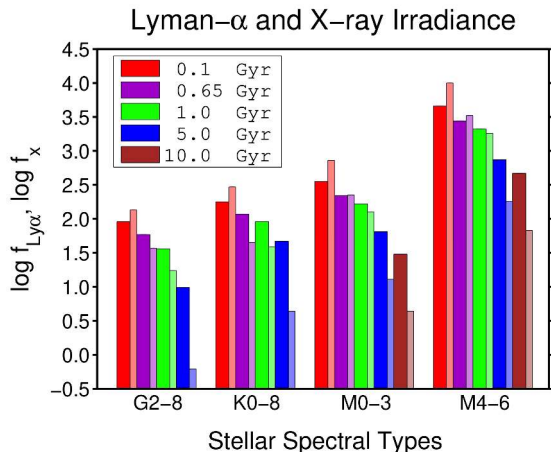
$$t_{\text{MS}} \approx 10 (M_{\star}/M_{\odot})^{-2.5} \text{ Gyr}$$

- $0.7 M_{\odot}$ K dwarf: ≈ 24 Gyr.
- $0.5 M_{\odot}$ early M: ≈ 57 Gyr.
- $0.1 M_{\odot}$ late M: ≈ 3000 Gyr.

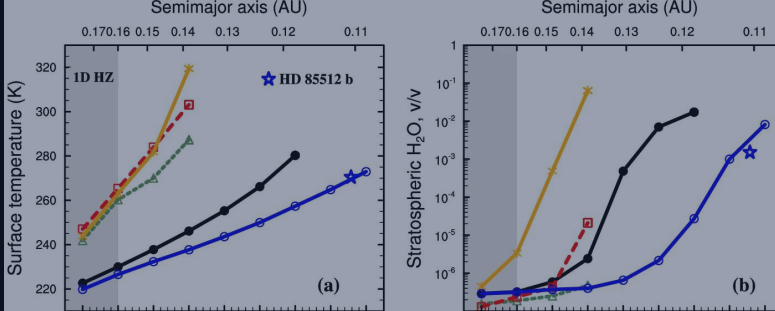
Cooler stars give biology more time.

$L \propto M^{3.5}$ is a single-power-law approximation; piecewise relations give $\sim M^4$ for $M \gtrsim 0.4 M_{\odot}$ and $\sim M^{2.3}$ below. Numbers good to factors of a few.

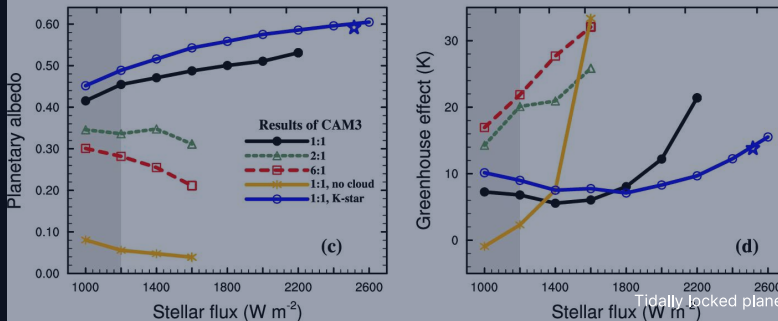
XUV at the HZ: M Dwarfs are Hostile



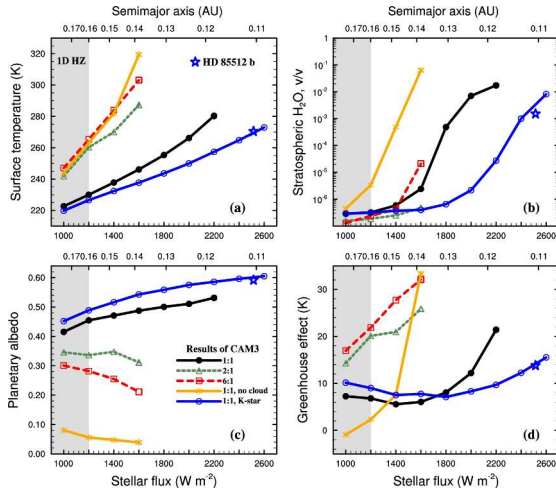
- Lyman- α (wide bars) + X-ray (narrow bars) at the Earth-equivalent distance, by spectral type and age.
- At fixed age, $> 100\text{--}500\times$ higher from G2-8 to M4-6.
- X-rays drop by up to $500\times$ from 0.1 to 5 Gyr; Lyman- α only $\sim 5\text{--}10\times$.
- Early K: the compromise. Long-lived and common, without the M dwarf extremes.
- Cuntz & Guinan (2016).



6. HZ Corrections: 3D Climate, Tidal Locking

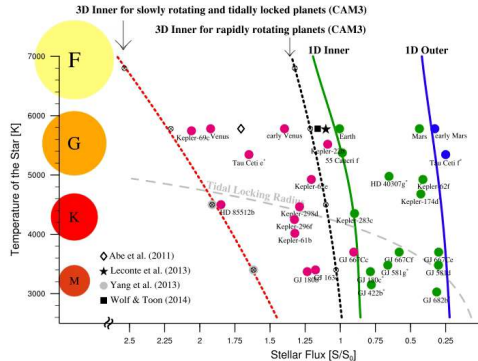


3D GCMs Shift the Inner Edge of the M Dwarf HZ



- Yang+ 2013: tidally locked planet GCM (global climate model) with clouds.
- Substellar convection lifts thick water clouds.
- Albedo rises to ~ 0.6 .
- Stabilising feedback against runaway.
- Inner edge tolerates $\sim 2\times$ higher stellar flux.
- 1D HZ underestimates M dwarf habitable region.

Where M Dwarf HZ Planets Actually Sit



- Shields+ 2016. Flux increases to the *left*.
- Green/blue solid: 1D Kopparapu inner, outer edges.
- Dotted: 3D inner edge, red for slow and locked, black for rapid rotators.
- Red is at $1.7\text{--}2.2\times$ the 1D inner-edge flux.
- Grey dashed: tidal-locking radius. Nearly all M dwarf HZ planets are on or inside it.

Tidal-Locking Timescale

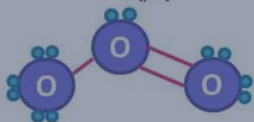
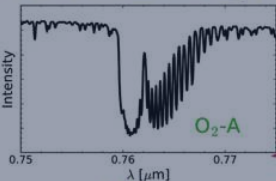
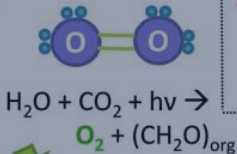
$$\tau_{\text{lock}} \sim \frac{\omega_0 a^6 I_p Q_p}{3 G M_{\star}^2 k_{2,p} R_p^5}$$

Strong a^6 dependence:

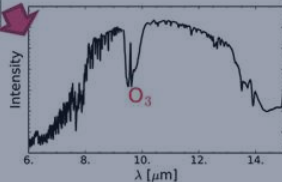
- Earth at 1 AU: $\tau_{\text{lock}} \sim 10^{12}$ yr (much longer than universe age).
- TRAPPIST-1 b ($a \approx 0.011$ AU): $\sim 10^7$ yr or shorter.
- TRAPPIST-1 system age: ~ 8 Gyr.
- Every M dwarf HZ planet has had time to lock.

Gaseous

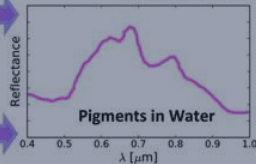
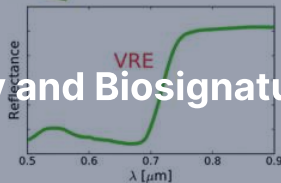
Ex: **Oxygenic Photosynthesis**



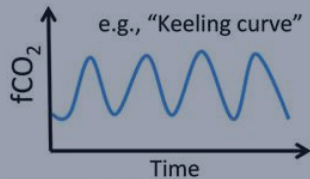
7. Astrobiology and Biosignatures



Surface



Temporal



Seasonal
Changes in
Gases or
Surface



What is Life?

Working definition (Joyce 1994, building on Lederberg 1965 + Viking-era debate):

- A self-replicating, metabolising, evolving chemical system.
- Operational, not philosophically deepest.
- Lets us decide what to look for.

Chemical baseline:

- Carbon-based, water-as-solvent.
- Only example we have.
- Dominates the search because we know its spectroscopic + geochemical fingerprints.

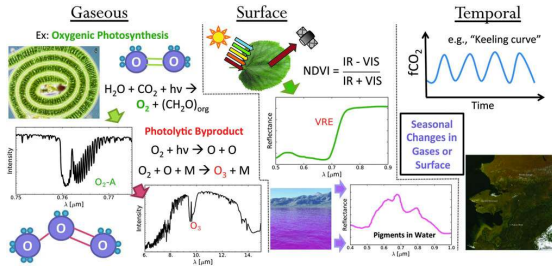
Origin of Life: Four Live Hypotheses

Not solved. Present as competing, not as a settled story.

- **RNA world:** ribozyme (catalytic RNA)-driven, RNA precedes proteins. Hard prebiotic chemistry of RNA.
- **Metabolism-first / alkaline hydrothermal vents:** autocatalytic cycles precede information storage. Geochemistry compatible.
- **Surface "warm little ponds":** wet-dry cycling concentrates and polymerises. Darwin 1871; revived by experiment.
- **Panspermia:** relocates the problem, doesn't solve it.

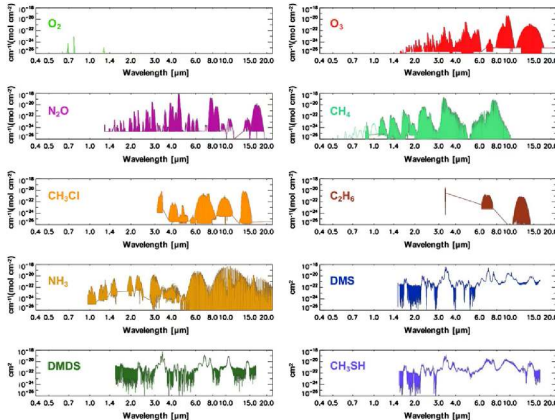
Earliest direct evidence on Earth: ~ 3.5 Ga. Possible to ~ 4.0 Ga (contested).

Three Classes of Biosignature



- **Gaseous:** O_2 , O_3 , CH_4 , N_2O , organosulphur, organohalogen.
- **Surface:** vegetation red edge (sharp reflectance jump) near 700 nm.
- **Temporal:** seasonal cycles, e.g. Earth's CO_2 Keeling curve.
- Convincing claim should combine ≥ 2 classes.
- Schwieterman+ (2018).

Biosignature Gases in the Mid-IR



- Ten potential biosignature gases.
- Each has a characteristic IR absorption signature.
- O₃: 9.6 μm. CH₄: 3.3 + 7.7 μm. N₂O: across mid-IR.
- DMS / DMDS: 3–15 μm window.
- Drives the LIFE / HWO wavelength designs.

Disequilibrium is the Diagnostic

Single gas in isolation: rarely convincing.

Most diagnostic: *disequilibrium combinations*.

- Earth: O_2 (21%) + CH_4 (1.8 ppm). React on ~decade timescales.
- Joint presence requires continuous biological resupply (oxygenic photosynthesis + methanogenesis).
- Pre-Great Oxidation Event (before O_2 rose, ~2.4 Ga) Earth: CH_4 + N_2O or CH_4 + CO_2 .
- Krissansen-Totton+ 2018 formalised disequilibrium approach.

Catling 2018: A Bayesian Framework

EXOPLANET BIOSIGNATURES ASSESSMENT

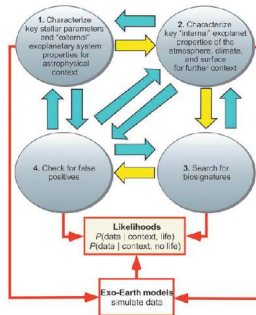
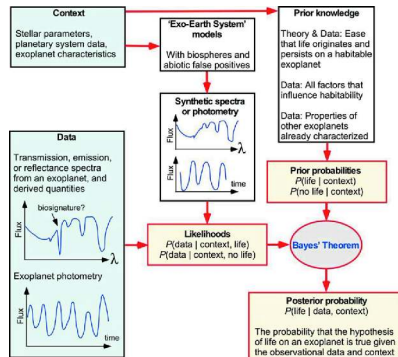
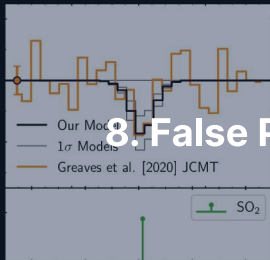


FIG. 2. Exo components to assess whether extant



Detection is a quantitative inference from a chain of measurements: priors on host star, planet, atmosphere, false-positive pathways. Not a yes/no statement on a single observation.

8. False Positives and Solar System Targets



False Positives: Every Gas Has an Abiotic Pathway

- **Abiotic O_2** : H_2O photolysis (light-driven splitting) + H escape (Wordsworth+ 2014, Luger & Barnes 2015). CO_2 photolysis.
- **Abiotic CH_4** : serpentinisation (H_2 -releasing reaction of olivine with water). Volcanic outgassing at low f_{O_2} .
- **Abiotic N_2O** : lightning chemistry, photochemistry.
- **Abiotic DMS**: no significant Earth source known. But "not on Earth" $\neq$ "impossible elsewhere".

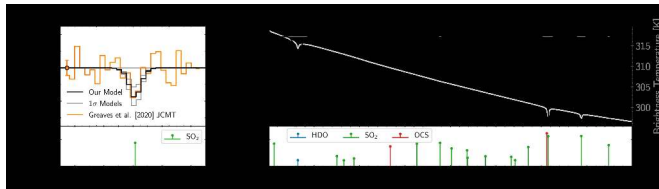
Biosignature detection is an inverse problem: must rule out *all* plausible abiotic pathways.

Solar System Targets for Life Detection

Five solar-system bodies under active astrobiological assessment:

- **Mars:** past habitability established. Mars Sample Return is the path forward.
- **Europa:** confirmed subsurface ocean. Clipper at Jupiter 2030.
- **Enceladus:** plumes deliver ocean chemistry (H_2 + organics + phosphates) to your spacecraft. Decadal-Survey priority.
- **Titan:** complete hydrocarbon hydrology + transient impact-melt water. Dragonfly arrives 2034.
- **Venus cloud layer:** phosphine claim contested.

Venus Phosphine: A Cautionary Case Study

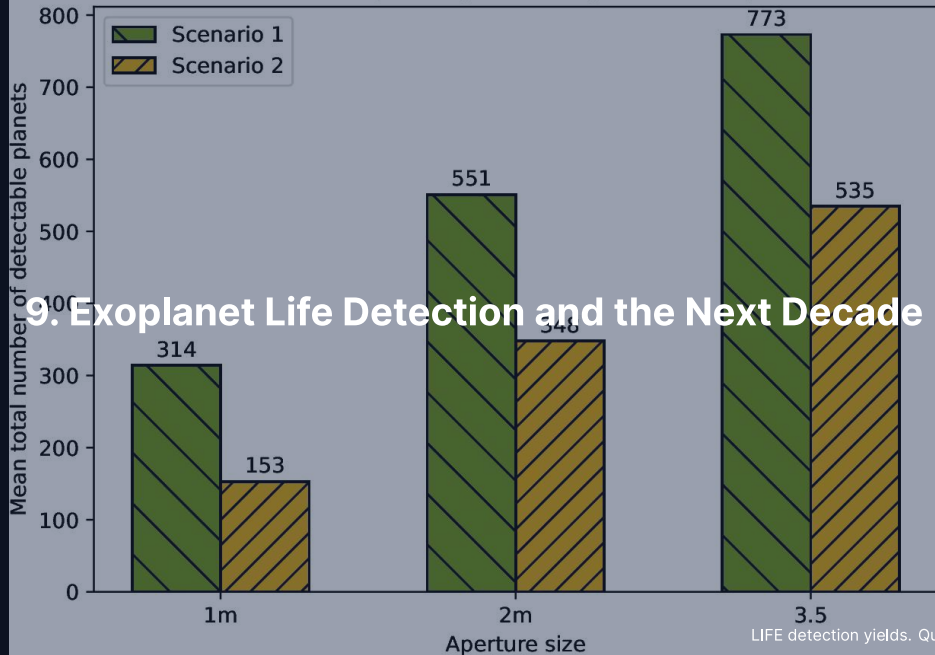


- Greaves+ 2021: PH₃ in Venus clouds claimed as biosignature.
- Lincowski+ 2021: same data fits SO₂ in Venus mesosphere.
- Original feature is at the JCMT/ALMA noise floor.
- Direct analogue of the K2-18 b DMS controversy.
- DAVINCI + EnVision + VERITAS will resolve.

Venus Phosphine + K2-18 b: Same Lesson, Two Telescopes

- Both: data at edge of instrumental sensitivity.
- Both: molecular identification contested.
- Both: plausible abiotic explanations not ruled out.
- Right response in either case: *better data*, ideally from independent instruments.
- Both episodes have driven the design of next-generation instruments.
- Pedagogical value is high regardless of how each turns out.

Total LIFE exoplanet yield (2.5-year search phase)

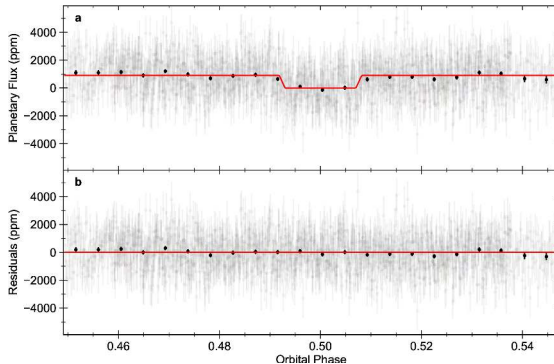


9. Exoplanet Life Detection and the Next Decade

What Convincing Exoplanet Biosignature Detection Requires

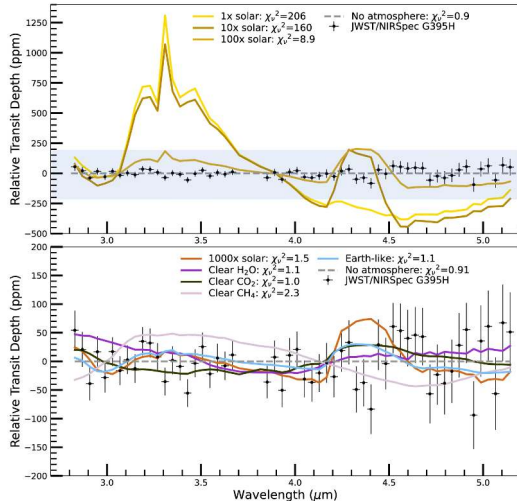
1. Multiple independent gases in disequilibrium (oxidising + reducing).
2. Temporal variability consistent with a biological cycle.
3. Geological context (mass, radius, host, age, escape) excluding obvious abiotic pathways.
4. Independent confirmation by different instrument / wavelength / observatory.

TRAPPIST-1 b: What JWST Can Already Tell Us



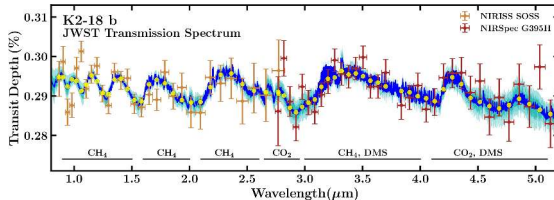
- MIRI 15 μm secondary eclipse.
- $T_d \approx 508 \text{ K}$ = bare-rock equilibrium.
- No thick atmosphere to redistribute heat.
- Greene+ 2023.
- M dwarf HZ planets often retain little atmosphere.

LHS 475 b: Featureless Spectrum



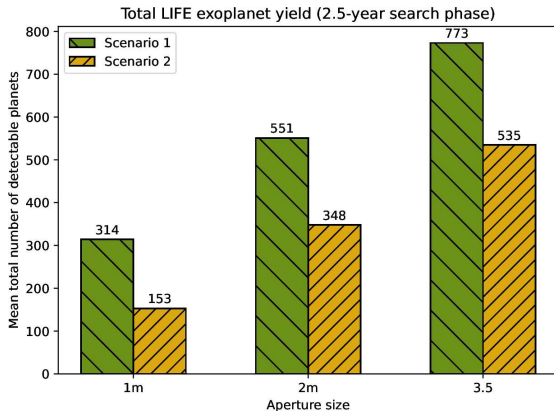
- NIRSpec G395H, Earth-sized M dwarf planet.
- Featureless transmission.
- H_2 -dominated ruled out.
- High-mean-molecular-weight (CO_2 -rich) or airless: still consistent.
- Lustig-Yaeger+ 2023.

K2-18 b: How Tentative Biosignatures Are Tested



- Madhusudhan+ 2023: CH₄ at $\sim 5\sigma$, CO₂ at $\sim 3\sigma$.
- DMS only 2.4 σ , and $\sim 1\sigma$ once one instrument offset floats.
- Hycean (H₂ over global ocean) interpretation contested.
- Multiple groups failed to reproduce DMS.
- Whatever the resolution: **textbook case in real-time biosignature testing.**

LIFE: Mid-IR Nulling Interferometer



- Four free-flying spacecraft, 4–18 μm .
- Nulling: stellar light suppressed by destructive interference.
- 3.5 m aperture: 500–800 planets, ~ 25 rocky HZ.
- Quanz+ 2022.
- Companion to HWO (visible/NIR).

The 2026–2040 Mission Queue

- **Now–2030:** JWST continues. BepiColombo at Mercury 2026. PLATO 2026. Roman 2027. Dragonfly launch 2028. Ariel 2029. Europa Clipper at Jupiter 2030.
- **2030s:** JUICE Ganymede orbit. Dragonfly at Titan 2034. Ariel atmospheric statistics. ELT class first light 2028+. DAVINCI/EnVision/VERITAS at Venus. Mars Sample Return delivery (re-baselined). Uranus orbiter concept advances.
- **2040s:** HWO + LIFE in hardware. First plausible direct atmospheric characterisation of *Earth-analogue* exoplanets.

The first time “are we alone?” is empirical, not philosophical. The empirical regime begins within the careers of the students in this room.

The Five Biggest Things We Have Learned

1. Planet formation is a **physical process**, not chance. Same laws everywhere.
2. Planetary interiors are **heat engines** driving tectonics, dynamos, atmospheres on Gyr timescales.
3. Atmospheres are **not static**. They form, evolve, escape, and feed back on surfaces and biospheres.
4. Habitability is a **coupled systems property**, not a single condition.
5. The solar system is **one example** in a much larger statistical population.

The Five Biggest Open Questions

1. How does life originate, and how often?
2. What sets the radius valley (gap between super-Earths and sub-Neptunes)?
3. When did Jupiter form, and was it the architect of the inner solar system?
4. Was Mars ever inhabited?
5. Is the solar system rare or typical?

Final Framing

- Planetary science is now *the science of comparative climate, interior, and life-hosting trajectories*.
- Solar system: reference, not benchmark.
- Exoplanet population: statistical context.
- The frontier is moving fast.
- One systems-level view: habitability, formation, escape, weathering, biology are *one coupled problem*, with one consistent set of physical principles, the same set wherever you go in the galaxy.

That is what the course has been about.



Thank You

Planetary Systems is complete.

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience

